

LEGAL AI COPILOT

The AI Operations Copilot for Law Firms & Legal Teams

Turn client chats, emails, and legal documents into safe, cited, automated legal workflows.

Document purpose. This PDF defines product and system requirements for the Legal AI Copilot using FlowOps-style sections (problem framing → MVP scope → architecture → agents/tools → security & ops).

Primary outcomes

- Answer client questions with a consistent safe format (Summary → Legal basis → Next steps → Disclaimer).
- Ground answers in Egyptian legal sources via Retrieval-Augmented Generation (RAG) with citations.
- Automate lawyer/admin workflows (reminders, scheduling, follow-ups) using n8n/Zapier + webhooks.
- Provide auditability, role-based access control, and observability for enterprise use.

Audience: product, engineering, and delivery teams building the capstone system.

What the System Is

A legal copilot that supports end-to-end legal work: client intake and Q&A, research over Egyptian law, document drafting, and automated task execution. The system combines an LLM backend with RAG, an agent orchestrator, and workflow automation.

Core capabilities

- Client chat (“Ask AI Lawyer”) with safe response template and clarifying questions when needed.
- RAG over Egyptian legal sources (laws, amendments, regulations, judgments, books) with citations.
- Lawyer drafting support (contracts, pleadings, motions) grounded in retrieved sources and templates.
- Automation for reminders, scheduling, follow-ups, and daily digests via integration workflows.
- Evaluation and observability for quality, safety, latency, and cost.

Who Pays / Primary Users

Target customers are law firms and legal departments that need faster intake, research, drafting, and deadline tracking, with enterprise controls and auditability.

Role	Primary goals	Key permissions (examples)
Client	Ask questions, submit documents, track status	Ask AI Lawyer; upload files; view own matter updates
Lawyer	Research, draft, manage matters and deadlines	RAG search; drafting; create tasks/reminders; approve outgoing actions
Office Head / Partner	Oversight, KPIs, risk management	View dashboards; approve policies; access audit logs; manage templates
Admin / Paralegal	Intake, scheduling, coordination	Create/update matters; schedule events; send notifications with approval rules

MVP Scope (Capstone Modules)

Module 1 — Ask AI Lawyer (Client Chat)

- Provide a consistent response format: Summary → Legal basis → Next steps → Disclaimer.
- Ask clarifying questions when the user input is incomplete or ambiguous.
- Use RAG for legal grounding and enforce citations when retrieval is used.
- Apply safety guardrails: refuse disallowed requests, detect prompt injection, and warn/redact sensitive data (PII).
- Support human handoff when legal risk is high or confidence is low.

Module 2 — Workflow Automation (n8n/Zapier + Webhooks)

- Create reminders and scheduling events for deadlines, hearings, and follow-ups.
- Generate daily/weekly digests for each matter (tasks due, pending approvals, new uploads).
- Extract structured fields from emails/WhatsApp/notes (parties, dates, claim type, requested action).
- Require human approval for actions that contact external parties (emails/messages).
- Provide retries, dead-letter handling, and visible failure reporting.

Module 3 — Lawyer Copilot (RAG + Agents)

- RAG over Egyptian legal sources with versioning/effective dates and citation-ready snippets.
- Hybrid retrieval (keyword + vectors) with reranking and deduplication to improve relevance.
- Draft legal documents using templates and retrieved citations; clearly state assumptions.
- Quality checks: missing clauses, contradictions, risky language, and formatting rules.
- Evaluation suite for RAG faithfulness, citation correctness, and agent tool-call success.

Definition of Done (MVP)

- End-to-end demo: client asks a question, system retrieves sources, answers with citations, then schedules a follow-up via automation.
- Role-based access works (client cannot access other matters; staff actions logged).
- RAG returns citations with correct source identifiers and offsets (article/section/page).
- Basic observability dashboard exists (latency, cost, retrieval success, errors).

System Architecture (High-Level)

The system follows a layered architecture: Web UI → API Gateway → Copilot Services (chat, RAG, agents) → Tools/Integrations → LLMs and data stores.

Layer	Responsibilities	Key components
Web UI	Client/lawyer portal, file upload, matter view, approvals	Chat UI; admin console
API Gateway	Auth, rate limiting, routing, request validation	JWT/OAuth; WAF; request logging
Copilot Services	Chat, orchestration, safety, state, formatting	Conversation svc; policy engine; structured output
RAG Service	Ingestion, chunking, retrieval, citation packaging	Indexer; vector DB; BM25; reranker
Agent Orchestrator	Planning, tool calls, multi-agent collaboration	Planner; worker agents; tool router
Tools & Integrations	Calendars, email, workflow automation, case mgmt	n8n/Zapier; webhooks; connectors
Data Stores	Matters, users, audit logs, docs, embeddings	Relational DB; object store; log store
LLM Providers	Reasoning, drafting, summarization, routing	Primary LLM; fallback models; safety filters

Key non-functional targets (guidance)

- Latency: interactive chat under a few seconds for non-retrieval answers; retrieval answers may be slower but must stream progress.
- Reliability: idempotent tool calls; retrievable automation with failure visibility.
- Security: least-privilege access, encrypted storage in transit/at rest, complete audit trail.
- Compliance: handle client data safely; provide configurable retention and redaction policies.

RAG (Legal Brain)

The Legal Brain provides citation-backed retrieval over Egyptian legal sources and internal firm knowledge (templates, playbooks, FAQs).

- **Data sources:** Egyptian laws, amendments, executive regulations, judgments, legal books, and firm templates.
- **Ingestion:** scheduled sync, deduplication, metadata enrichment (jurisdiction, effective date, article number).
- **Chunking:** structure-aware chunking (articles/sections), preserve headings and numbering.
- **Retrieval:** hybrid search (BM25 + vectors) plus reranking; return top-k with confidence signals.
- **Citations:** every grounded statement links to source id + location (article/section/page).
- **Safety:** prompt-injection scanning on retrieved text and uploaded files; strip/neutralize instructions in sources.

RAG acceptance tests

- Given a question about an article of law, the system retrieves the correct article and cites it.
- If sources disagree or are outdated, the answer highlights the conflict and asks for context (date, jurisdiction).
- If retrieval is empty or low-confidence, the system explicitly says it cannot confirm and suggests next steps.

Agent Design (Gold-tier)

Use a planner-and-workers approach: an orchestrator decomposes tasks, invokes tools, and merges results. Each worker agent has a narrow responsibility and clear inputs/outputs.

Agent	Responsibilities	Typical outputs
Planner / Orchestrator	Decompose tasks, route to tools/agents, enforce safety and formatting	Plan; tool calls; merged answer
Research Agent	RAG retrieval, build citation pack, summarize sources	Citation pack; summaries
Drafting Agent	Generate drafts grounded in citations and templates	Drafts with citations
Review Agent	Check completeness, contradictions, risky language, formatting	Issues + edits
Automation Agent	Schedule reminders, create digests, execute approved actions	Events; tasks; triggers
Reporting Agent	Matter updates and dashboards	Status + KPIs

Agent safety rules

- Agents must never fabricate citations; if retrieval is missing, they must say so.
- Tool calls require a clear, logged rationale and validated parameters.
- External communications require approval unless explicitly configured otherwise.

Tools (Function Calling Requirements)

All tool calls must be auditable, idempotent where possible, and recorded with inputs, outputs, and the approving user (if applicable).

Tool	Purpose	Key requirements
rag_search(query, filters)	Retrieve passages and citations	Structured citations; confidence; block injection patterns
generate_document(type, inputs)	Draft legal documents from templates	Template selection; versioning; citations + assumptions embedded
extract_case_fields(text/files)	Extract parties, dates, issues, tasks	Schema validation; PII handling; confidence + missing fields
create_calendar_event(details)	Schedule deadlines/hearings	Timezone-safe; dedupe/idempotency; attach matter id
create_reminder(details)	Deadline reminders and follow-ups	Escalation rules; retry; notify via configured channel
send_message(draft)	Email/WhatsApp/SMS (approved)	Approval gate; rate limits; full audit trail
log_action(event)	Audit log entry	Append-only; immutable ids; exportable reports

Multimodal Inputs

The system must accept and reason over text plus uploaded files (PDF/DOCX/images) and messages (email/WhatsApp), while enforcing safety and privacy policies.

- Upload pipeline: virus scan (if available), file type validation, OCR optional, and safe text extraction.
- Prompt injection detection on document content; treat docs as data, not instructions.
- PII handling: detect and redact sensitive identifiers in logs and optional in answers.
- Source attribution: when the model uses a document, it must cite it and include the relevant excerpt location.

Security (Enterprise-grade)

Security requirements focus on client confidentiality, least privilege, guardrails against misuse, and comprehensive auditability.

- **Auth & access control:** JWT/OAuth; role-based access; matter-based authorization checks on every request.
- **Data protection:** encryption in transit and at rest; secrets stored in a vault; configurable retention policies.
- **Guardrails:** prompt-injection defenses; refusal policies; safe completion policies for legal advice.
- **Audit trail:** log all chat prompts (redacted), retrieval sources, tool calls, approvals, and system actions.
- **Rate limiting & abuse prevention:** per-user quotas; anomaly detection; lockouts for repeated violations.

Security acceptance tests

- A client cannot view or retrieve any content from another matter.
- A tool call to send an external message cannot execute without an approval record (unless policy explicitly allows).
- Audit logs are immutable (append-only) and exportable for review.

Observability, Evaluation, and Optimization

The system must be measurable and improvable: capture traces and metrics, evaluate RAG and drafting quality, and optimize cost/latency while maintaining safety.

- **Observability:** request traces, tool-call traces, retrieval metrics (hit rate, top-k quality), error logs with PII redaction.
- **Evaluation:** offline test set for Egyptian law Q&A; citation correctness checks; regression tests for tool-call flows.
- **Feedback loop:** user thumbs-up/down, issue tagging, and data collection for prompt/retrieval tuning.
- **Optimization:** caching, context trimming, model routing, and retrieval tuning (chunk size, reranker thresholds).

Release checklist

- All modules runnable via docker-compose; environment variables documented.
- Minimum monitoring in place (latency, errors, cost) with alerts for failures.
- Evaluation report produced for baseline quality and citation correctness.